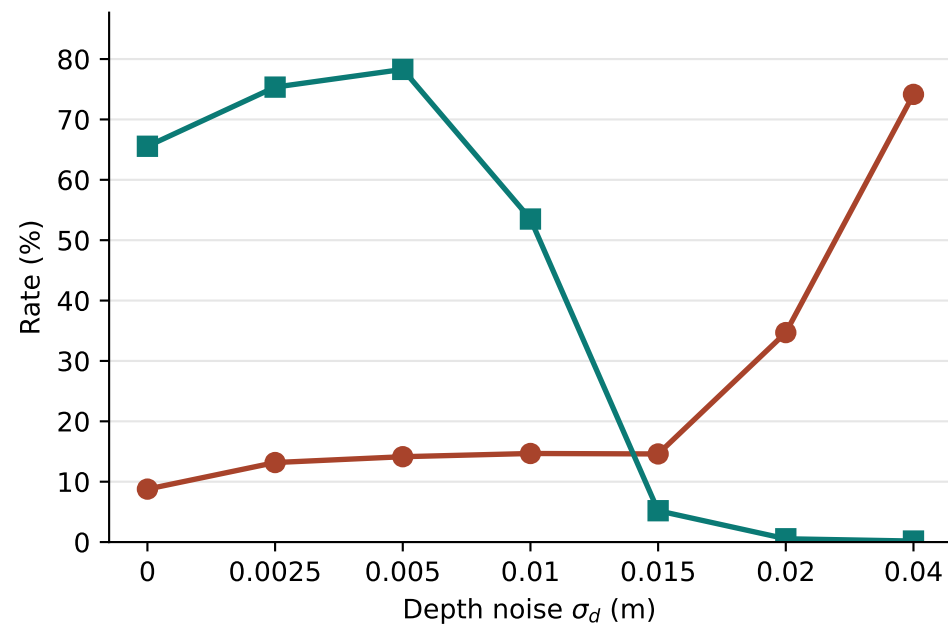
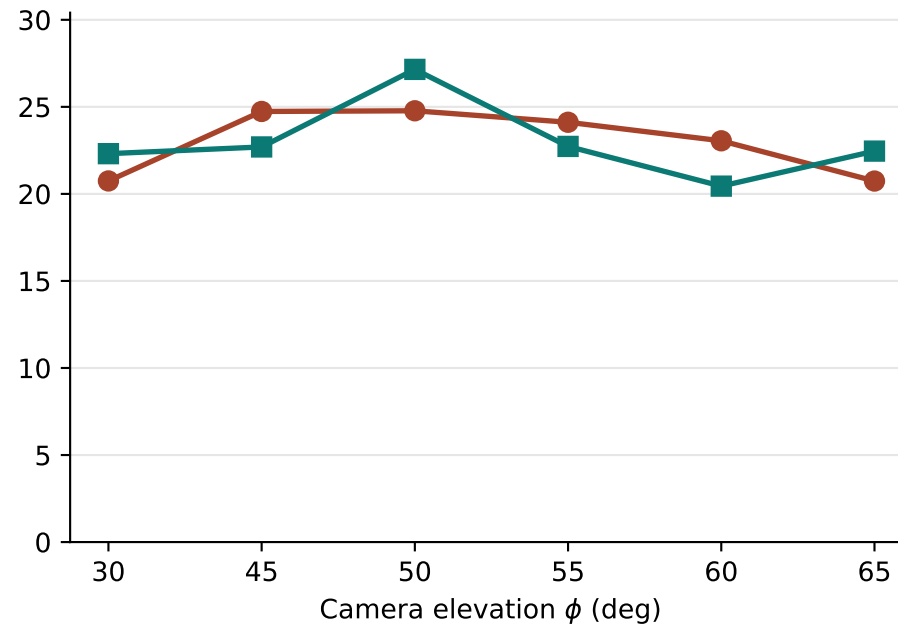


Grasp failure decomposes into a geometric gate and an execution channel ($n = 7,560$ trials, floating gripper, top-1 pose)

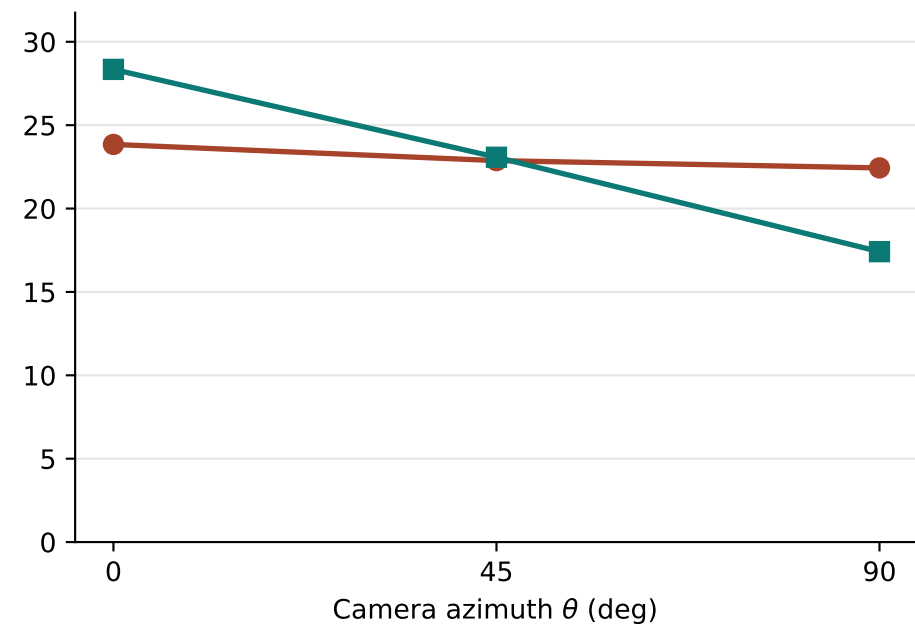
(a) Depth noise moves the two channels in opposite directions



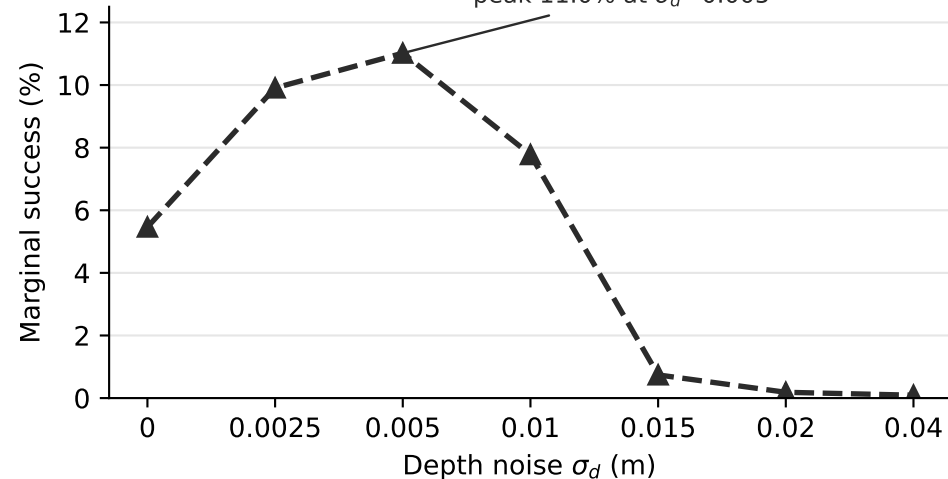
(b) Elevation moves neither channel



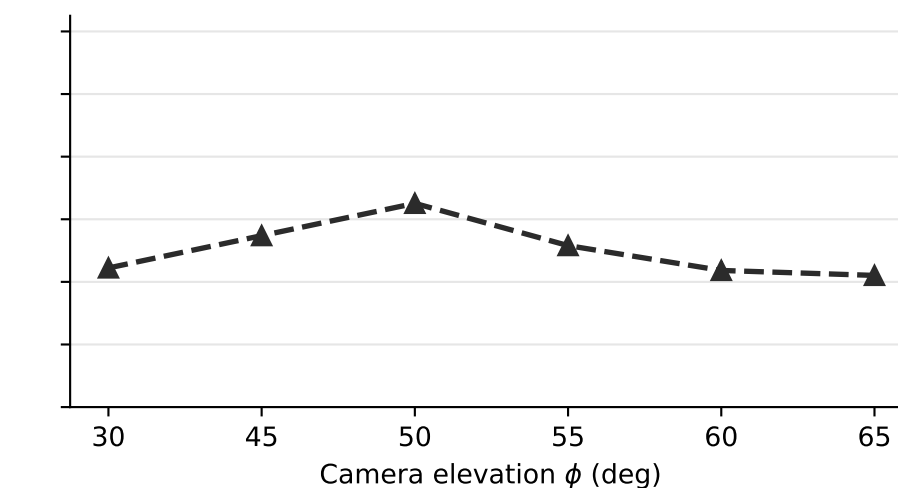
(c) Azimuth acts on execution only (pooled; see object split)



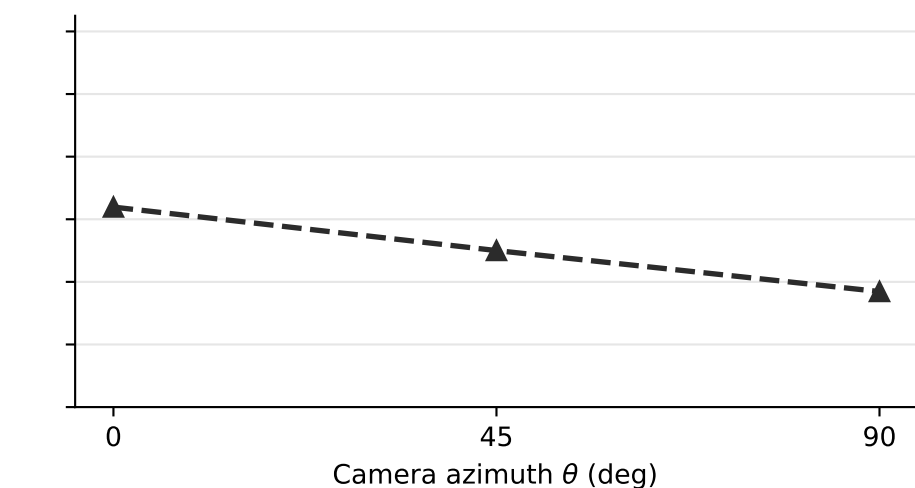
(d) Marginal $P(Y=1)$ against σ_d
peak 11.0% at $\sigma_d=0.005$



(e) against ϕ



(f) against θ



—●— Pose clears open-hand collision check —■— Lift succeeds | pose cleared check —▲— Marginal success $P(Y=1)$